

(normative)

Maintenance plan guidelines

The purpose of this annex is to provide a template to guide the preparation of a software maintenance plan based on this standard.

The maintenance plan (MP) should contain the content as described in C.1 through C.8. The user of this annex may adopt any format and numbering system for the MP. The MP section numbers listed in this annex are provided to assist in the readability of this annex and are not mandatory for the user.

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Figure C.1—Example software maintenance plan outline

C.1 (MP Section 1) Introduction

The MP should describe the specific purpose, goals, and scope of the software maintenance effort, including deviations from this standard. The software maintenance effort for which the plan is being written and the specific software processes and products covered by the software maintenance effort should be identified. Date of plan issue and status should be provided. Plan issuing organization and approval authority should be identified. (See A.3.1.)

C.2 (MP Section 2) References

The MP should identify the documents placing constraints on the maintenance effort, documents referenced by the MP, and any supporting documents supplementing or implementing the MP including other plans or task descriptions that elaborate details of this plan.

C.3 (MP Section 3) Definitions

The MP should define or reference all terms required to understand the MP. All abbreviations and notations used in the MP should be described.

C.4 (MP Section 4) Software maintenance overview

The MP should describe organization, scheduling priorities, resources, responsibilities, tools, techniques, and methods necessary to perform the software maintenance process. (See A.3, A.4, A.5, A.12, and Annex B.)

C.4.1 (MP Section 4.1) Organization

The MP should describe the organization of the software maintenance effort. The MP should describe the lines of communication with the software maintenance effort including external organizations, the authority for resolving issues raised in the software maintenance effort, and the authority for approving software maintenance products.

C.4.2 (MP Section 4.2) Scheduling priorities

The MP should describe how the maintenance activity will be grouped into work packages, the factors that determine the organizational maintenance priorities, and the process for assigning a priority to a work package and how the resources are assigned to prioritized work packages. The schedule estimating method should be described.

C.4.3 (MP Section 4.3) Resource summary

The MP should summarize the software maintenance resources, including staffing, facilities, tools, finances, and special procedural requirements (e.g., security, access rights, and documentation control). The cost estimating method should be described.

C.4.4 (MP Section 4.4) Responsibilities

The MP should identify an overview of the organizational element(s) and responsibilities for maintenance activities.

C.4.5 (MP Section 4.5) Tools, techniques, and methods

The MP should describe the special documents, software maintenance tools, techniques, methods, and operating and test environment to be used in the maintenance process. Acquisition, training, support, and qualification information for each tool, technology, and methodology should be included. The MP should document the measurements and metrics to be used by the maintenance process and should describe how these measurements and metrics support the maintenance process.

C.5 (MP Section 5) Software maintenance process

The MP should identify actions to be performed for each of the software maintenance phases described in Clause 4, and should document those actions. The MP should contain an overview of the maintenance phases. (See 4.1 through 4.7, and A.4.)

C.5.1 (MP Sections 5.1 through 5.7) Software maintenance process

The MP should include sections 5.1 through 5.7 for software maintenance phases as shown in the MP outline (see Figure C.1).

The MP shall address the following topics for each software maintenance phase:

- a) *Phase input.* What is needed to perform the phase.
- b) *Phase output.* What results when the phase is performed.
- c) *Phase process.* The details of what a phase is expected to do.
- d) *Phase controls.* What is to be performed to control the results of the phase.

NOTE—The user of this template should examine 4.1 through 4.7 for process details.

C.6 (MP Section 6) Software maintenance reporting requirements

The MP should describe how information will be collected and provided for each reporting period, including: work packages completed, work packages in-work, work packages received, and backlog. Also, risks should be identified along with their mitigation approach. (See A.7, A.8, and A.11.)

C.7 (MP Section 7) Software maintenance control requirements

The MP should describe the anomaly resolution and reporting; deviation policy; control procedures; and standards, practices, and conventions. (See A.11.)

C.7.1 (MP Section 7.1) Anomaly resolution and reporting

The MP should describe the method of reporting and resolving anomalies, including the criteria for reporting an anomaly, the anomaly distribution list, and authority for resolving anomalies.

C.7.2 (MP Section 7.2) Deviation policy

The MP should describe the procedures and forms used to deviate from the plan. The MP should identify the authorities responsible for approving deviations.

C.7.3 (MP Section 7.3) Control procedures

The MP should identify control procedures applied during the maintenance effort. These procedures should describe how software products and maintenance results should be configured, protected, and stored.

C.7.4 (MP Section 7.4) Standards, practices, and conventions

The MP should identify the standards, practices, and conventions that govern the performance of maintenance actions including internal organizational standards, practices, and policies.

C.7.5 (MP Section 7.5) Performance tracking

The MP should describe the procedures for tracking performance through all software maintenance phases for each work item.

C.7.6 (MP Section 7.6) Quality control of plan

The MP should describe how the plan is reviewed, updated, and approved to ensure plan correctness and currency.

C.8 (MP Section 8) Software maintenance documentation requirements

The MP should describe the procedures to be followed in recording and presenting the outputs of the maintenance process as specified in 4.1.4, 4.2.4, 4.3.4, 4.4.4, 4.5.4, 4.6.4, and 4.7.4. (See A.5.)